ECEN 749 Lab 4 - Report

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Introduction

This lab focuses on booting Linux on a Zybo Z7-10 board using an SD card. The process involves constructing a Zynq-based microprocessor system capable of running Linux. Subsequently, the bitstream is integrated with the First Stage Boot Loader and Universal Boot Loader to generate the Zynq boot image. These combined files are then utilized to facilitate the Linux boot process on the Zybo board.

Procedure

1. Begin by creating a new block design. Integrate the Zynq7 processing system IP and utilize block automation to import the Zybo_Z7_B2.tcl file. Recustomize the IP to enable only the SD 0, UART 1, and TTC 0 peripherals. Incorporate the multiply IP from the previous lab's IP repository into the block design. Subsequently, run connection automation, generate the HDL wrapper and bitstream, and export the hardware design as an .xsa file.

2. Petalinux installation should be done in the /tmp directory with a folder petalinux_netid. The petalinux installer is first copied from the shared path to the current directory and it is then installed. After installing this, a new directory called linux_boot is created where a petalinux project of template Zynq is created and in this directory the previously created .xsa file needs to be reconfigured. After changing the branch from master to main in SRC_URI in the cracklib_2.9.8.bb file petalinux-build can be executed. The next step is to generate the Zynq boot image (BOOT.bin). This will generate 3 files: the BOOT.bin, image.ub and the boot.scr which will all have to be copied onto the SD card. Change the jumper position to SD card, run picocom and on pressing the PR-SRST linux will start booting on
the board which can be observed on the terminal.

Results

The Linux operating system's boot process was observed via the picocom console, which displayed a user login and password prompt. To validate the OS functionality, root access was obtained using sudo su, and a test file named "hello world" was created and opened successfully.

Conclusion

This lab provided valuable experience in creating a custom hardware multiplication IP and developing the corresponding C code for peripheral testing. The C code verified the IP's functionality by writing to two registers and then reading them using the m_ReadReg statement. We also gained practical knowledge by interpreting commands from header files to implement

the multiplication IP. Furthermore, we learned to locate registers using base addresses and offsets, and ultimately observed the multiplier in action via the picocom console.

Questions

a) Linux uses 512 MB of SDRAM. CPU cache is the system's local memory however registers are even more local to the system than the cache and registers are the fastest accessible to the CPU. It ensures data access and temporary file storage. It is included in all mother boards in the Zynq Zybo board; The cache acts as a mediator between the SDRAM and registers.

b) Only the proc and sys folders are read-only; all other folders are writable. Files created in these writable folders using the touch command are temporary. They are not permanently stored to the SD card due to permission settings and will be lost upon board restart. Therefore, these files must be recreated after each restart for continued use.

c) Integrating a new peripheral into a system after kernel compilation necessitates a re-execution of the configuration and compilation processes. This involves modifying the device tree configuration to accommodate the requisite drivers and hardware parameters. The initial step is to update the hardware design by incorporating the new peripheral into the block design, generating the bitstream, and exporting the hardware design (.xsa) file. Subsequently, Petalinux tools are used to reconfigure the project with the updated .xsa file. Upon successful generation of the BOOT.bin, image.ub, and boot.scr files, these are then copied to the SD card to initiate the operating system on the Zybo board.

Appendix

Picocom console:

```
FATAL: read zero bytes from port
term_exitfunc: reset failed for dev UNKNOWN: Input/output error
yashas1729@zach-333-em4-13:~$ picocom -b 115200 /dev/ttyUSB1
picocom v3.1

port is        : /dev/ttyUSB1
flowcontrol    : none
baudrate is    : 115200
parity is      : none
databits are   : 8
stopbits are   : 1
escape is      : C-a
local echo is  : no
noinit is      : no
noreset is     : no
hangup is      : no
nolock is      : no
send_cmd is    : sz -vv
receive_cmd is : rz -vv -E
imap is        :
omap is        :
emap is        : crcrlf,delbs,
logfile is     : none
initstring     : none
exit_after is  : not set
exit is        : no

Type [C-a] [C-h] to see available commands
Terminal ready
ext3: Unknown parameter 'umask'
ext2: Unknown parameter 'umask'
ext4: Unknown parameter 'umask'
FAT-fs (mmcblk0p1): Volume was not properly unmounted. Some data may be corrupt. Please run fsck.
hwclock: can't open '/dev/misc/rtc': No such file or directory
Fri Mar  9 12:34:56 UTC 2018
hwclock: can't open '/dev/misc/rtc': No such file or directory
random: crng init done
Configuring packages on first boot....
 (This may take several minutes. Please do not power off the machine.)
Running postinst /etc/rpm-postinsts/100-sysvinit-inittab...
update-rc.d: /etc/init.d/run-postinsts exists during rc.d purge (continuing)
 Removing any system startup links for run-postinsts ...
  /etc/rcS.d/S99run-postinsts
INIT: Entering runlevel: 5
Configuring network interfaces... Cannot find device "eth0"
```

```
ext2: Unknown parameter 'umask'
ext4: Unknown parameter 'umask'
FAT-fs (mmcblk0p1): Volume was not properly unmounted. Some data may be corrupt. Please run fsck.
hwclock: can't open '/dev/misc/rtc': No such file or directory
Fri Mar  9 12:34:56 UTC 2018
hwclock: can't open '/dev/misc/rtc': No such file or directory
random: crng init done
Configuring packages on first boot....
 (This may take several minutes. Please do not power off the machine.)
Running postinst /etc/rpm-postinsts/100-sysvinit-inittab...
update-rc.d: /etc/init.d/run-postinsts exists during rc.d purge (continuing)
 Removing any system startup links for run-postinsts ...
  /etc/rcS.d/S99run-postinsts
INIT: Entering runlevel: 5
Configuring network interfaces... Cannot find device "eth0"
Starting OpenBSD Secure Shell server: sshd
  generating ssh RSA host key...
  generating ssh ECDSA host key...
  generating ssh ED25519 host key...
done.
Starting rpcbind daemon...done.
starting statd: done
hwclock: can't open '/dev/misc/rtc': No such file or directory
Starting internet superserver: inetd.
NFS daemon support not enabled in kernel
Starting syslogd/klogd: done
Starting tcf-agent: OK

PetaLinux 2023.1+release-S05010539 linux_boot ttyPS0

linux_boot login: petalinux
You are required to change your password immediately (administrator enforced).
New password:
Retype new password:
linux_boot:~$ sudo su

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

Password:
linux_boot:/home/petalinux#
```